Predictive Classification of Credit Card Fraud Using Random Forest Ensembles

Dataset: creditcard.csv (https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud)

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1. Introduction

Credit card fraud is a serious problem because fraudulent transactions can cause financial loss to customers and banks. This project uses a Random Forest classification model to classify transactions as legitimate (Class 0) or fraudulent (Class 1).

2. Objective

The objective is to apply a Random Forest ensemble to a real-world classification dataset, prepare the data, train a model, and evaluate it using the metrics required in the minor-project guidelines.

3. Dataset Description

The provided creditcard.csv file contains 284,807 transactions and 31 columns. The input columns are Time, V1–V28, and Amount. The target column is Class. Class 0 represents legitimate transactions and Class 1 represents fraud. The dataset contains 284,315 legitimate transactions and 492 fraud transactions. No missing values were found during inspection.

4. Data Preparation

The original dataset is inspected first. Because fraud cases are very rare and the full dataset is relatively large, a reproducible stratified sample of 60,000 records is used for the model section to keep execution manageable. The sample preserves approximately the same class proportions. The target Class column is separated from the input features. The data is divided into 80% training data and 20% testing data using stratification, so both sets retain the rare fraud class.

5. Model Development

A RandomForestClassifier is used. The model uses 40 trees (n_estimators=40), a maximum tree depth of 10 (max_depth=10), and class_weight='balanced_subsample' to give additional consideration to the minority class within each tree's sample. random_state=42 makes the result reproducible.

6. Evaluation Results

The following results were obtained from the 60,000-row stratified sample, using an 80:20 train-test split. Values are rounded to four decimal places.

Metric	Result
Accuracy	0.9992 (99.92%)
Precision (fraud class)	0.8667
Recall (fraud class)	0.6190
F1-score (fraud class)	0.7222

ROC-AUC	0.9258

Confusion Matrix

	Predicted legitimate	Predicted fraud
Actual legitimate	11,977	2
Actual fraud	8	13

Interpretation: 13 fraud transactions were correctly identified, 8 fraud transactions were missed, and 2 legitimate transactions were incorrectly flagged as fraud in the test set.

7. Discussion

The model's accuracy is high, but accuracy should not be considered on its own because the dataset is strongly imbalanced. Precision indicates how many predicted fraud cases were actually fraud. Recall indicates how many actual fraud cases were found. F1-score balances precision and recall. ROC-AUC measures how well the model separates the two classes across possible decision thresholds. The confusion matrix shows the counts of correct and incorrect predictions.

8. Limitations

The model is trained on a sample rather than all 284,807 rows to reduce training time. A larger training run may produce different results. V1–V28 are anonymized features and their original meanings are not available in this dataset. The model is a learning project and should not be used as a real banking fraud-detection system without further testing and validation.

9. Conclusion

This project demonstrates a complete basic classification workflow using Random Forest: dataset inspection, class-imbalance analysis, train-test splitting, model training, and evaluation. The results show why recall, precision, F1-score, and ROC-AUC are important for fraud detection. Future improvements could include testing more hyperparameter settings, adjusting the classification threshold, and comparing the model with other classifiers.